

Review of Wilsonian Renormalization

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1 Motivating the Beta Functions[1]

We start with a rather different problem from classical dynamics. Consider a particle moving according to the “Aristotle’s law”,

$$\frac{dx(t)}{dt} = -\frac{dV(x)}{dx} \quad (1)$$

$V(x)$ can be thought of as a terrain in one dimension. Let the terrain has a maximum at x_c and two local minima at $x_a < x_c$ and $x_b > x_c$ ¹. The differential equation can be solved to get $x(t; x_0)$. Depending upon the initial condition x_0 , the position at some time t will be different. Given infinite time, $x(\infty; x_c - \epsilon) = x_a$ and $x(\infty; x_c + \epsilon) = x_b$. There is a singular behavior, in $x(\infty; x_0)$. However for any finite t , $x(t; x_0)$ is a continuous function of x_0 (a plot is shown below).

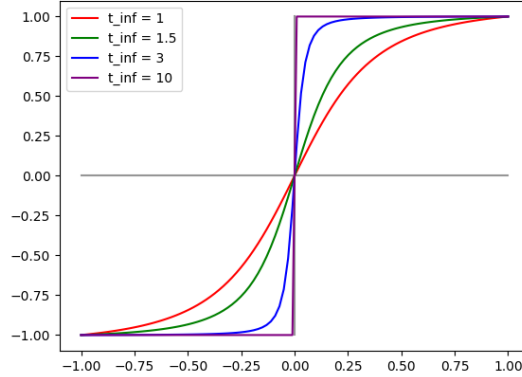


Figure 1: Numerical solution for $V(x) = x^4/4 - x^2/2$ potential

Even from an analytic function $V(x)$, we got some singular behaviour out, when we allowed certain parameter in the system go to infinity. Something on this similar line happens in the physics of critical phenomena. The ising model partition function is given by,

$$Z(K, h) = \sum_{\{s_i\}} \exp \left\{ K \sum_{\langle ij \rangle} s_i s_j + h \sum_i s_i \right\} \quad (2)$$

where $K = -\frac{J}{T}$ and $h = -\frac{B}{T}$; J is the nearest neighbor coupling constant, B is the external magnetic field, T is the temperature, with Boltzmann’s constant taken to be 1.

¹A potential of the form $-x^2 + x^4$ will do the job

This computation involves exponential function of K and h , which is analytic and hence no reason to suspect a singularity in Z . But, when we calculate the free energy taking the infinite volume limit,

$$F(K, h) = \lim_{V \rightarrow \infty} \frac{1}{V} \ln Z(K, h) \quad (3)$$

we see certain singular behaviour in various quantities, marked by universal values known as critical exponents. Wilson showed that such singularities can be explained using differential equations analogous to the rolling ball problem.

Kadanoff earlier proposed a scheme to retain the scaling relations obeyed by the free energy and correlation length, using some ad hoc hypothesis. Wilson uses this scheme as a toy model to motivate the construction of the differential equations of the Renormalization Group(RG), which are now called the *Beta functions*.

The Kadanoff scaling procedure is as follows. We take the ising system of d dimensions and make d dimensional cubic compartments in it (called blocks), with each block having a side with b spin sites. Hence, any block will have b^d spins in it. He assumes that all spins in one block are either “up” or “down” (which is a bit dodgy). Then define a new spin variable, s'_i which is the total sum of the block, normalized. Since all spins in a block is assumed to be the same, the new spin variable also take values ± 1 . Kadanoff further assumes that the Hamiltonian after the “blocking” procedure will be of the same form, but with the new spin variables and couplings denoted by $K(b)$ and $h(b)$. This means that $Z(K(b), h(b)) = Z(K, h)$ which would have been more reasonable if we didn’t assume all spins of a block aligned up. Now,

$$F(K(b), h(b)) = \lim_{V_b \rightarrow \infty} \frac{1}{V_b} \ln Z(K(b), h(b)) = \lim_{V \rightarrow \infty} \frac{b^d}{V} \ln Z(K, h) = b^d F(K, h) \quad (4)$$

Since the blocking procedure changes the unit of measurement of length from 1 spin as a unit to b spins as a unit, the correlation length will obey the following,

$$\xi(K(b), h(b)) = b^{-1} \xi(K, h) \quad (5)$$

Now, in the original Kadanoff picture, it makes sense to choose b as an integer. But, Wilson assumes that b takes continuous values between 0 and ∞ . The beta functions then are defined as the differential equations for $K(b)$ and $h(b)$ as b is changing.

Now, notice that K_{2b} is a function of only $K(b)$ and $h(b)$ but does not explicitly depend on b . This is because we can think of $K(b), h(b)$ represent the “original” lattice, and then doing a scaling by making new blocks with each side having ℓ number of old blocks to get to $K_{\ell+b}$ and $h_{\ell+b}$. This new couplings cannot have a memory of the b . Hence, K_{nb} and h_{nb} will be only a function of only $K(b)$ and $h(b)$. Take a special $n = 1 + \delta$, for a small δ . Using $K_{(1+\delta)b} = K(b) + \delta b \frac{dK(b)}{db}$, we can say that $\delta b \frac{dK(b)}{db} \propto$ (a function of $K(b)$ and $h(b)$). Hence we write,

$$\frac{dK(b)}{db} = \frac{1}{b} u(K(b), h(b)^2) \quad (6)$$

Similarly,

$$\frac{dh(b)}{db} = \frac{1}{b} h(b) v(K(b), h(b)^2) \quad (7)$$

The functional dependence on $h(b)$ is taken as $h(b)^2$ because we have the symmetry $K \rightarrow K$ and $h \rightarrow -h$ in the ising model. So K must be an even function of $h(b)$, and h should be an odd function of $h(b)$, because of which we have explicitly put one $h(b)$ factor in the right hand side of the equation for $h(b)$. Now, equation 6 and equation 7 are the Beta functions we are looking for. We make one more additional assumption. Just like we had $V(x)$ analytic for the rolling ball problem, we assume u and v are analytic functions.

2 The Scaling Hypothesis Derived[1]

If we integrate the differential equations from $b = 1$ to $b \sim \xi \equiv \xi(K, h)$, then from equation 5, we can say that $\xi(K(b), h(b)) \sim 1$. For this to happen, the values of $K(b)$ and $h(b)$ should be well away from the critical point,

denoted by K_c and $h_c = 0$. Let us choose an initial condition, in which $K < K_c$ (we are above the critical temperature). Now if we integrate the differential equations with larger and larger b , $K(b)$ has to increase to go away from the critical value (otherwise the correlation length won't decrease). If the initial conditions were exactly at the critical point, then no matter what b is, $K(b) = K_c$ and $h(b) = 0$. For these to happen, we need,

$$u(K_c, 0) = 0 \quad (8)$$

and no restrictions to v , thanks to the $h(b)$ factor multiplying v in equation 7. Now assume we start off with values of K and h , really close to the critical point. Then for small values of b , $\xi(K(b), h(b)) \sim \xi$, which is large, and hence $K(b)$ and $h(b)$ are still near the critical point. Therefore, we can linearise the differential equations near the critical point for small b . Thus we have,

$$\frac{dK(b)}{db} \simeq \frac{K(b) - K_c}{b} y \quad \text{where} \quad y = \frac{\partial u(K_c, 0)}{\partial K(b)} \quad (9)$$

and

$$\frac{dh(b)}{db} \simeq \frac{h(b)}{b} x \quad \text{where} \quad x = v(K_c, 0) \quad (10)$$

Integrate this equation till a small value of b ,

$$\ln(K(b) - K_c) \Big|_1^b = y \ln b \Big|_1^b \Rightarrow K(b) = K_c + \epsilon b^y \quad (11)$$

where $\epsilon = K - K_c$. Likewise,

$$\ln(h(b)) \Big|_1^b = x \ln b \Big|_1^b \Rightarrow h(b) = h b^x \quad (12)$$

The process of linearising the equation is equivalent to using a quadratic potential near x_c in the rolling ball problem. If we use the approximation till a value of $K(b)$ well away from the critical value, say $K_c/2$, then we can solve for the value of b for which $K(b) = K_c/2$,

$$b = \left(\frac{K_c}{2\epsilon} \right)^{1/y} \quad (13)$$

For this particular b , we have,

$$h(b) = h \left(\frac{K_c}{2\epsilon} \right)^{x/y} \quad (14)$$

Then using equation 4, we get,

$$F(K, h) = \left(\frac{K_c}{2\epsilon} \right)^{-d/y} F \left(K_c/2, h \left(\frac{K_c}{2\epsilon} \right)^{x/y} \right) \quad (15)$$

and,

$$\xi(K, h) = \left(\frac{K_c}{2\epsilon} \right)^{1/y} \xi \left(K_c/2, h \left(\frac{K_c}{2\epsilon} \right)^{x/y} \right) \quad (16)$$

The above two equations are the scaling hypothesis! Notice that the functional dependence on the right hand side is only on one variable, $h\epsilon^{-x/y}$

To be more accurate, we should have been solving the full non linear equation, when we are far away from the critical point. Let,

$$K(b) = \phi(b, h_0) \quad (17)$$

$$h(b) = \psi(b, h_0) \quad (18)$$

be the solution for the non linear equations, with the boundary conditions,

$$\phi(1, h_0) = \frac{K_c}{2} \quad (19)$$

$$\psi(1, h_0) = h_0 \quad (20)$$

Now, for very small values of b , these solutions should behave as the linearised solutions. Hence when $b \ll 1$,

$$\phi(b, h_0) \simeq \phi(h_0)b^y + K_c \quad (21)$$

$$\psi(b, h_0) \simeq \psi(h_0)b^x \quad (22)$$

where $\phi(h_0)$ and $\psi(h_0)$ are functions which depends on h_0 . Now, notice that the original differential equations equation 6 and equation 7 are translational invariant if we use $\ln b$ as the variable. This means that,

$$K(b) = \phi(\alpha b, h_0) \quad (23)$$

$$h(b) = \psi(\alpha b, h_0) \quad (24)$$

are also true, for some arbitrary α . Now when $\alpha b \ll 1$, we will have the linearised solutions,

$$\phi(b, h_0) \simeq \phi(h_0)(\alpha b)^y + K_c \quad (25)$$

$$\psi(b, h_0) \simeq \psi(h_0)(\alpha b)^x \quad (26)$$

Comparing these with equation 11 and equation 12, we get,

$$\epsilon = \phi(h_0)\alpha^y \quad (27)$$

$$h = \psi(h_0)\alpha^x \quad (28)$$

Then usign these, we can find,

$$h\epsilon^{-x/y} = \psi(h_0)\alpha^x(\phi(h_0)\alpha^y)^{-x/y} = \frac{\psi(h_0)}{(\phi(h_0))^{x/y}} \quad (29)$$

This is an implicit equation for h_0 , which imply h_0 can depend only on $h\epsilon^{-x/y}$. Now, in this new setup, $K(b) = \frac{K_c}{2}$ when $b = a^{-1}$. Then using equation 4 and equation 5, we get,

$$F(K, h) = \left(\frac{\epsilon}{\phi(h_0)} \right)^{d/y} F\left(\frac{K_c}{2}, h_0 \right) \quad (30)$$

and

$$\xi(K, h) = \left(\frac{\epsilon}{\phi(h_0)} \right)^{-1/y} \xi\left(\frac{K_c}{2}, h_0 \right) \quad (31)$$

These are the scaling hypothesis relations again!

3 A Three Parameter Example[1]

Consider the following equations,

$$\begin{aligned} \frac{dK(b)}{db} &= \frac{1}{b} u(K(b), q(b), h(b)) \\ \frac{dq(b)}{db} &= \frac{1}{b} w(K(b), q(b), h(b)) \\ \frac{dh(b)}{db} &= \frac{1}{b} h(b) v(K(b), q(b), h(b)) \end{aligned} \quad (32)$$

The initial values for $b = 1$ will be denoted by K, q and h . The relations for the Free energy and the correlation function are assumed to remain the same.

$$F(K, q, h) = b^{-d} F(K(b), q(b), h(b)) \quad (33)$$

$$\xi(K, q, h) = b \xi(K(b), q(b), h(b)) \quad (34)$$

For a fixed value of q , the problem is exactly the same as that we had previously. There will be an equilibrium point $K_c(q)$ and $h = 0$ for each q . Hence as q changes, we expect a line of critical points. If we start with a point on the line, the values for the couplings will stay on the line, no matter what the value of b is. This is because on each point on the line, correlation length is infinite, and if we do not stay on the line, ξ won't remain infinite. On the critical line, we have,

$$\frac{dq(b)}{db} = \frac{1}{b} w(K_c(q(b)), q(b), 0) = \frac{1}{b} w_c(q(b)) \quad (35)$$

The solution for the above equation can have two possible outcomes. One is that the value of $q(b)$ converges to a critical value, say q_c as b is increased. Or, $q(b)$ may run off to infinity. Let $K_c(q_c) = K_c$. Then $(K_c, q_c, 0)$ is an equilibrium point for the entire system of differential equations in equation 32. Now, if we start near the equilibrium point, we can linearise the system. We have the following two conditions to ensure that once we start at the equilibrium point it stays there,

$$u(K_c, q_c, 0) = 0 \quad \text{and} \quad w(K_c, q_c, 0) = 0 \quad (36)$$

The linearise equations are,

$$b \frac{dK(b)}{db} = y_{11}(K(b) - K_c) + y_{12}(q(b) - q_c) \quad (37)$$

$$b \frac{dq(b)}{db} = y_{21}(K(b) - K_c) + y_{22}(q(b) - q_c) \quad (38)$$

$$b \frac{dh(b)}{db} = xh(b) \quad (39)$$

The solutions should also obey a power law,

$$K(b) - K_c = b^y \quad q(b) - q_c = r_y b^y \quad (40)$$

and

$$K(b) - K_c = b^z \quad q(b) - q_c = r_z b^z \quad (41)$$

There are two possibilities near the critical point as b increases; either go to the equilibrium point, or run off to infinity. Consider the case with $z < 0$ and $y > 0$. This makes the point K_c, q_c a saddle point, attractive in one direction and repulsive in another. The general solution will be,

$$K(b) = K_c - \epsilon b^y + \eta b^z \quad (42)$$

$$q(b) = q_c - r_y \epsilon b^y + r_z \eta b^z \quad (43)$$

$$h(b) = h b^x \quad (44)$$

Using an initial condition $K(0) = K$ and $q(0) = q$, we can solve for the values of ϵ and η ,

$$\epsilon = \frac{r_z(K - K_c) - (q - q_c)}{r_y - r_z} \quad \text{and} \quad \eta = \frac{r_y(K - K_c) - (q - q_c)}{r_y - r_z} \quad (45)$$

The critical line then would be described by $\epsilon = 0, h = 0$ and $\eta \neq 0$. Anywhere on this line, as $b \rightarrow \infty$, the values approach K_c, q_c .

When $\epsilon \ll \eta$ and a small h , we are near the critical line. When $b \sim 1$, the η term dominates and the point flows towards the fixed point. But when b becomes large, the ϵ term dominates, hence the flow will be away from the

fixed point. This flow pattern is independent of the starting value of η . Now, if we get the value of b for which $K(b)$ becomes $K_c/2$, (such a b would be large, hence only the ϵ term plays a role)

$$\frac{K_c}{2} \simeq \epsilon^y \quad (46)$$

Which gives,

$$b \simeq \left(\frac{K_c}{2\epsilon} \right)^{1/y} \quad (47)$$

This gives,

$$q(b) \simeq q_c - r_y \left(\frac{K_c}{2\epsilon} \right) \quad \text{and} \quad h(b) \simeq h \left(\frac{K_c}{2\epsilon} \right)^{x/y} \quad (48)$$

The value of $q(b)$ for this particular value of b is just a constant, independent of ϵ, η or h . Hence the scaling relations are exactly as the same as we had for the two variable example.

The same kind of analysis can be carried out for $T < T_c$. We cannot find the value of b for which $K(b)$ becomes $K_c/2$ in this case. This is easily remedied by finding the value of b for which $K(b)$ becomes a larger value than K_c , say $2K_c$.

When $z < 0$ and $y > 0$, we can get to the fixed point by tuning two initial parameters. But if we consider the case when both y and z are positive, then K_c, q_c is still a critical point, but unstable in all directions. So to sit on top of that point, we have to tune all three parameters.

4 The Ising Model and Critical Exponents[2]

Now, we apply the ideas discussed above to the Ising model. For this, we will use the Renormalization group procedure described by Wilson, in his second paper, but present it in a modern notation. The general Ising model Hamiltonian in d dimensions is given by,

$$H = -\frac{1}{2} \sum_{i,j} J_{ij} s_i s_j - B \sum_i s_i \quad (49)$$

Where J_{ij} is the interaction term between spins at sites i and j , and B is the external magnetic field applied. We are interested in finding the partition function, given by,

$$Z = \sum_{\{s_i\}} \exp \left\{ \frac{\beta}{2} \sum_{i,j} J_{ij} s_i s_j + \beta B \sum_i s_i \right\} = \sum_{\mathbf{s}} \exp \left\{ \frac{1}{2} \mathbf{s}^T \tilde{J} \mathbf{s} + \tilde{\mathbf{h}}^T \mathbf{s} \right\} \quad (50)$$

Where $\mathbf{s}$ and $\tilde{\mathbf{h}}$ are column vectors and $\tilde{J}$ is an $N \times N$ matrix. All factors of β are taken care of in the definitions of $\tilde{\mathbf{h}}$ and $\tilde{J}$. It can be shown that the Ising model can be cast into a functional integral form[2],

$$Z = \frac{\int D\phi \exp\{-S[\phi]\}}{\int D\phi \exp\left\{-\frac{1}{2} \phi^T \tilde{J} \phi\right\}} = \sqrt{\det \tilde{J}} \int D\phi \exp\{-S[\phi]\} \quad (51)$$

where,

$$S[\phi] = \frac{\beta}{2} \sum_{i,j} J_{ij} \phi_i \phi_j - \sum_{i=1}^N \ln \left\{ 2 \cosh \left[\beta \left(h + \sum_{j=1}^N J_{ij} \phi_j \right) \right] \right\} \quad (52)$$

where ϕ is the spin equivalent variable with ϕ_i having an expectation value $\langle s_i \rangle$. Now, take a Fourier transform,

$$\phi_i = \sum_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{r}_i} \phi_{\mathbf{k}} \quad (53)$$

with the sum is taken in the range $0 \leq |\mathbf{k}| < \frac{1}{a}$, where a is the lattice spacing. Assuming the fluctuations from the mean is small, we can expand out cosh term as a power series, using,

$$\ln [2 \cosh x] = \ln 2 + \frac{x^2}{2} - \frac{x^4}{12} + \dots \quad (54)$$

we get,

$$S[\phi] = -N \ln 2 + \frac{\beta}{2} \sum_{i,j} J_{ij} \phi_i \phi_j - \frac{\beta^2}{2} \sum_{i=1}^N \left(h + \sum_{j=1}^N J_{ij} \phi_j \right)^2 + \frac{\beta^4}{12} \sum_{i=1}^N \left(h + \sum_{j=1}^N J_{ij} \phi_j \right)^4 + \dots \quad (55)$$

Now applying the Fourier transform, we can show that,

$$S[\phi] = -N \ln 2 - \beta^2 \sqrt{N} h J_{\mathbf{k}=0} \phi_{\mathbf{k}=0} + \frac{\beta}{2} \sum_{vbk} J_{\mathbf{k}} (1 - \beta J_{\mathbf{k}}) \phi_{\mathbf{k}} \phi_{-\mathbf{k}} + \frac{\beta^4}{12N} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} \delta_{\sum_i \mathbf{k}_i, 0} \prod_{i=1}^4 J_{\mathbf{k}_i} \phi_{\mathbf{k}_i} \quad (56)$$

where

$$J_{\mathbf{k}} = \sum_{i=1}^N e^{i\mathbf{k} \cdot \mathbf{r}_i} J(\mathbf{r}_i) \quad (57)$$

with $J_{ij} = J(\mathbf{r}_i - \mathbf{r}_j)$. Near the critical point, long range fluctuations, or equivalently, small $\mathbf{k}$ modes become dominant. Hence we can expand $J_{\mathbf{k}}$ can be expanded as a power series. Odd terms in $\mathbf{k}$ will not appear.² Then for nearest neighbour coupling,

$$J_{\mathbf{k}} = J(z - \mathbf{k}^2 a^2) \quad (58)$$

where z is the coordination number in d dimensions. Now taking the infinite volume limit, $Na^d \rightarrow \infty$, we can effectively replace $\frac{1}{V} \sum_{\mathbf{k}} \rightarrow \int \frac{d^d k}{(2\pi)^d}$ and define $\varphi(\mathbf{k}) = a\sqrt{V} \phi_{\mathbf{k}}$, we get,

$$S_{\Lambda_0}[\varphi] = V f_0 - h_0 \varphi(\mathbf{k}=0) + \frac{1}{2} \int \frac{d^d k}{(2\pi)^d} (r_0 + c_0 \mathbf{k}^2) \varphi(-\mathbf{k}) \varphi(\mathbf{k}) + \frac{u_0}{4!} \int \left(\prod_{i=1}^4 \frac{d^d k_i}{(2\pi)^d} \right) (2\pi)^d \delta^d \left(\sum_{i=1}^4 \mathbf{k}_i \right) \varphi(\mathbf{k}_1) \varphi(\mathbf{k}_2) \varphi(\mathbf{k}_3) \varphi(\mathbf{k}_4) \quad (59)$$

with Λ_0 being the upper limit of the integral, where $|\mathbf{k}| < \Lambda_0 \ll a^{-1}$
Equivalently in real space,

$$S_{\Lambda_0}[\varphi] = \int d^d x \left[f_0 + \frac{r_0}{2} \varphi^2(\mathbf{x}) + \frac{c_0}{2} (\nabla \varphi(\mathbf{x}))^2 + \frac{u_0}{4!} \varphi^4(\mathbf{x}) - h_0 \varphi(\mathbf{x}) \right] \quad (60)$$

5 Wilsonian Renormalization[3, 2, 4]

In this section, we describe Wilson's ideas on renormalization. It is carried out in two main steps. These steps, when described in words are deceptively simple, yet can be dauntingly difficult to carry out in practice. Step one of the procedure is;

Decimation: In this step, we eliminate the the degrees of freedom which represent short distance scales or large momentum scales. As an example, consider the theory described by equation 59 with the external magnetic field is set to zero. Rewrite,

$$\varphi(\mathbf{k}) = \varphi^<(\mathbf{k}) + \varphi^>(\mathbf{k}) \quad (61)$$

where,

$$\varphi^<(\mathbf{k}) = \begin{cases} \varphi(\mathbf{k}) & \text{if } |\mathbf{k}| < \Lambda \\ 0 & \text{otherwise} \end{cases} \quad \text{and} \quad \varphi^>(\mathbf{k}) = \begin{cases} \varphi(\mathbf{k}) & \text{if } \Lambda < |\mathbf{k}| < \Lambda_0 \\ 0 & \text{if } |\mathbf{k}| > \Lambda_0 \end{cases} \quad (62)$$

²since J_{ij} is real, $J_{\mathbf{k}} = J_{-\mathbf{k}}$

here, $\Lambda = \frac{\Lambda_0}{b}$ for some dimensionless number $b > 0$. Denote all of the coupling in $S_{\Lambda_0}[\varphi]$ by a vector $\mathbf{g}$. We then integrate out the higher momentum modes to get;

$$Z = \int D\varphi^< \int \varphi^> e^{-S_{\Lambda_0}[\varphi^< + \varphi^>; \mathbf{g}]} = \int D\varphi^< e^{-S_{\Lambda}^<[\varphi^<; \mathbf{g}^<]} \quad (63)$$

where,

$$e^{-S_{\Lambda}^<[\varphi^<; \mathbf{g}^<]} = \int \varphi^> e^{-S_{\Lambda_0}[\varphi^< + \varphi^>; \mathbf{g}]} \quad (64)$$

The coupling constants $\mathbf{g}^<$ are in general different from what we started with. $S_{\Lambda}^<[\varphi^<; \mathbf{g}^<]$ is called the *Wilsonian Effective action*. Since the action in equation 59 is valid near the critical point, and near the critical point, long range correlations are observed, we expect that $S_{\Lambda}^<[\varphi^<; \mathbf{g}^<]$ will have the same form as of $S_{\Lambda_0}[\varphi; \mathbf{g}]$.

Hence,

$$\begin{aligned} S_{\Lambda}^<[\varphi^<; \mathbf{g}^<] &= V f^< + \frac{1}{2} \int \frac{d^d k}{(2\pi)^d} (r^< + c^< \mathbf{k}^2) \varphi^<(-\mathbf{k}) \varphi^<(\mathbf{k}) \\ &+ \frac{u^<}{4!} \int \left(\prod_{i=1}^4 \frac{d^d k_i}{(2\pi)^d} \right) (2\pi)^d \delta^d \left(\sum_{i=1}^4 \mathbf{k}_i \right) \varphi^<(\mathbf{k}_1) \varphi^<(\mathbf{k}_2) \varphi^<(\mathbf{k}_3) \varphi^<(\mathbf{k}_4) + \text{terms involving } \phi^6, \phi^8 \dots \end{aligned} \quad (65)$$

Here, the upper limit of the momentum integral is taken to be Λ . Here ends the first step of the process. Integrating out the higher modes is usually impossible. We do this process perturbatively. Now, the second and final step of RG is;

Rescaling: We rescale the momentum variable,

$$\mathbf{k}' = b\mathbf{k} \quad (66)$$

and the field,

$$\varphi'(\mathbf{k}') = \zeta_b^{-1} \varphi^<(\mathbf{k}) \quad (67)$$

The field rescaling factor, $\zeta_b = b^{\Delta_{\varphi}} \sqrt{Z_b}$. Δ_{φ} is called the canonical dimension, which is obtained from pure dimensional analysis. Z_b is called the field strength renormalization factor, and it is chosen in such a way that the coefficient of the $\mathbf{k}^2$ term (c_0) remain unchanged after renormalization³,

$$Z_b = \frac{c_0}{c^<} \quad (68)$$

Hence after rescaling,

$$\frac{1}{2} \int^{\Lambda} \frac{d^d \mathbf{k}}{(2\pi)^d} (r^< + c^< \mathbf{k}^2) \varphi^<(-\mathbf{k}) \varphi^<(\mathbf{k}) = \frac{1}{2} \int^{\Lambda_0} \frac{d^d \mathbf{k}'}{(2\pi)^d} (r' + c' \mathbf{k}'^2) \varphi'(-\mathbf{k}') \varphi'(\mathbf{k}') \quad (69)$$

which gives,

$$\varphi^<(\mathbf{k}) = b^{1+d/2} \sqrt{\frac{c_0}{c^<}} \varphi'(\mathbf{k}') \quad (70)$$

$$r' = b^2 Z_b r^< \quad \text{and} \quad c' = c_0 \quad \text{and so on...} \quad (71)$$

These are the two steps of RG. In the next section, we discuss some important properties of the RG procedure.

³Choosing c_0 as the unchanged coefficient is a convention

6 Properties of RG [2]

Under RG, one set of coupling constants is mapped to a new set. Denote this by,

$$\mathbf{g}' = \mathcal{R}(b; \mathbf{g}) \quad (72)$$

In the parameter space of all possible values of coupling constants, the RG procedure generates a flow. This flow can end up at some fixed point, or run off to infinity. If it reaches a fixed point say $\mathbf{g}^*$, then, doing RG further won't take it anywhere else.

$$\mathbf{g}^* = \mathcal{R}(b; \mathbf{g}^*) \quad (73)$$

From the first section, we have learned that at the critical point, the correlation length becomes infinite. And equation 5 tells us that,

$$\xi(\mathbf{g}') = \frac{\xi(\mathbf{g})}{b} \quad (74)$$

Hence, at a fixed point,

$$\xi(\mathbf{g}^*) = \frac{\xi(\mathbf{g}^*)}{b} \quad \text{which means} \quad \xi(\mathbf{g}^*) = 0 \quad \text{or} \quad \infty \quad (75)$$

Hence, $\xi = 0$ fixed point is called a trivial fixed point, and $\xi = \infty$ fixed points are called critical fixed points. If there is a basin of points in the coupling constant space such that if we start anywhere from that basin we end up at the critical point after sufficient number of RG iterations, then this basin is called the critical surface corresponding to that critical fixed point. Any where in the critical surface, the correlation length remain infinite as we have seen in section 1.

Now, take a point in the coupling space, $\mathbf{g}$ which is close to the fixed point. Then,

$$\delta \mathbf{g}' = \mathbf{g}' - \mathbf{g}^* = \mathcal{R}(b; \mathbf{g}) - \mathcal{R}(b; \mathbf{g}^*) \quad (76)$$

Since $\delta \mathbf{g} = \mathbf{g} - \mathbf{g}^*$ is small, we linearise the RG flow equation around the critical fixed point. Then

$$\delta g'^\mu = \mathcal{R}^\mu(b; \mathbf{g}^*) + \frac{\partial \mathcal{R}^\mu(b; \mathbf{g}^*)}{\partial g^\nu} \delta g^\nu - \mathcal{R}^\mu(b; \mathbf{g}^*) = \frac{\partial \mathcal{R}^\mu(b; \mathbf{g}^*)}{\partial g^\nu} \delta g^\nu \quad (77)$$

The matrix $\frac{\partial \mathcal{R}^\mu(b; \mathbf{g}^*)}{\partial g^\nu}$ may not be diagonalisable. This means that the left and right eigenvectors are not the same. We need only the left eigenvector to proceed. Let,

$$v_\mu^i \frac{\partial \mathcal{R}^\mu(b; \mathbf{g}^*)}{\partial g^\nu} = \lambda^i(b) v_\nu^i \quad (78)$$

where i is the index on all eigenvectors. Now we project the change $\delta g^m u$ on to these eigen directions.

$$\delta g^\mu v_\mu^i = u^i \quad (79)$$

These u^i do not get mixed under the linearised RG flow. They just get scaled;

$$(u')^i = v_\mu^i (\delta g')^\mu = v_\mu^i \frac{\partial \mathcal{R}^\mu(b; \mathbf{g}^*)}{\partial g^\nu} \delta g^\nu = \lambda^i(b) v_\nu^i \delta g^\nu = \lambda^i(b) u^i \quad (80)$$

No summation over latin indices anywhere.

For this reason, u^i are called scaling variables. Now, since the RG equation obey,

$$\mathcal{R}(b; \mathcal{R}(b'; \mathbf{g})) = \mathcal{R}(bb'; \mathbf{g}) \quad (81)$$

The linearised flow also obey this relation. Therefore, the linearised matrix $\frac{\partial \mathcal{R}^\mu(b; \mathbf{g}^*)}{\partial g^\nu}$ for different values of b commutes. This means that eigenvectors are b independent and,

$$\lambda^i(b) \lambda^i(b') = \lambda^i(bb') \quad (82)$$

This is satisfied by,

$$\lambda^i(b) = b^{y_i} \quad (83)$$

The scaling variables transform under RG as,

$$u'^i = b^{y_i} u^i \quad (84)$$

Set $b = e^\ell$ and cast the above equation in differential form,

$$\frac{du^i}{d\ell} = y_i u^i \quad (85)$$

No summation over i . The fixed point is now, $u^i = 0$. We classify each of the scaling variables as follows,

1. $y_i > 0 :: u_i$ is a *relevant* direction
2. $y_i < 0 :: u_i$ is an *irrelevant* direction
3. $y_i = 0 :: u_i$ is a *marginal* direction, hence should take into account the higher order terms also.

We have seen how the scaling hypothesis can be derived from such differential equations.

7 Renormalization of ϕ^4 Theory[2, 4, 3]

In section 5, we have laid out the general procedure of RG. Now, we take up a concrete example to illustrate the process. Consider, the ϕ^4 theory with no external magnetic field.

$$S_{\Lambda_0}[\varphi] = \int d^d x \left[\frac{1}{2} (\nabla \varphi(\mathbf{x}))^2 + \frac{r_0}{2} \varphi^2(\mathbf{x}) + \frac{u_0}{4!} \varphi^4(\mathbf{x}) \right] \quad (86)$$

Here, we have omitted the f_0 part, because it is not relevant in the present discussion, and re-scaled c_0 to 1, since we will be keeping as a constant throughout the RG procedure. In the momentum space,

$$S_{\Lambda_0}[\varphi_{\mathbf{k}}] = \frac{1}{2} \int^{\Lambda_0} \frac{d^d k}{(2\pi)^d} (\mathbf{k}^2 + r_0) \varphi(-\mathbf{k}) \varphi(\mathbf{k}) + \frac{u_0}{4!} \int^{\Lambda_0} \left(\prod_{i=1}^4 \frac{d^d k_i}{(2\pi)^d} \right) (2\pi)^d \delta^d \left(\sum_{i=1}^4 \mathbf{k}_i \right) \varphi(\mathbf{k}_1) \varphi(\mathbf{k}_2) \varphi(\mathbf{k}_3) \varphi(\mathbf{k}_4) \quad (87)$$

We change the cutoff by, $\Lambda = \frac{\Lambda_0}{b}$. Now write,

$$\varphi_{\mathbf{k}} = \varphi_{\mathbf{k}}^< + \varphi_{\mathbf{k}}^> \quad (88)$$

Split the action,

$$S_{\Lambda_0}[\varphi_{\mathbf{k}}] = S_0[\varphi_{\mathbf{k}}^<] + S_0[\varphi_{\mathbf{k}}^>] + S_{\text{mix}}[\varphi_{\mathbf{k}}^<, \varphi_{\mathbf{k}}^>] \quad (89)$$

where S_0 contains the quadratic part, and S_{mix} contains the quartic part, and has exactly the same form as that of the quadratic part of equation 87. There won't be any term from the quadratic part which is a mix of $\varphi^<$ and $\varphi^>$ after we integrate out the higher momentum modes. We have chosen,

$$S_{\text{mix}}[\varphi] = \int d^d x (\varphi(x))^4 \quad (90)$$

Now,

$$e^{-S_{\Lambda}[\varphi_{\mathbf{k}}^<]} = e^{-S_0[\varphi_{\mathbf{k}}^<]} \int D\varphi_{\mathbf{k}}^> e^{-S_0[\varphi_{\mathbf{k}}^>]} e^{-S_{\text{mix}}[\varphi_{\mathbf{k}}^<, \varphi_{\mathbf{k}}^>]} = e^{-S_0[\varphi_{\mathbf{k}}^<]} \left\langle e^{-S_{\text{mix}}[\varphi_{\mathbf{k}}^<, \varphi_{\mathbf{k}}^>]} \right\rangle_{0,>} \quad (91)$$

here $\langle \cdot \rangle_{0,>}$ denotes the expectation value with respect to $e^{-S_0[\varphi_{\mathbf{k}}^>]}$. Take logarithm on both sides to get,

$$S_{\Lambda}[\varphi_{\mathbf{k}}^<] = S_0[\varphi_{\mathbf{k}}^<] - \ln \left\{ \left\langle e^{-S_{\text{mix}}[\varphi_{\mathbf{k}}^<, \varphi_{\mathbf{k}}^>]} \right\rangle_{0,>} \right\} \quad (92)$$

Now, we assume the value of u_0 is small, and we can evaluate the integral perturbatively. Then using,

$$e^{-x} = 1 - x + \frac{x^2}{2} + \dots \quad \text{and} \quad \ln(1+x) = x - \frac{x^2}{2} + \dots \quad (93)$$

for small x , we get,

$$S_\Lambda[\varphi_{\mathbf{k}}^<] = S_0[\varphi_{\mathbf{k}}^<] + \langle S_{\text{mix}} \rangle_{0,>} - \frac{1}{2} \left[\langle S_{\text{mix}}^2 \rangle_{0,>} - (\langle S_{\text{mix}} \rangle_{0,>})^2 \right] + \dots \quad (94)$$

First, we calculate $\langle S_{\text{mix}} \rangle_{0,>}$.

$$\begin{aligned} \langle S_{\text{mix}} \rangle_{0,>} = 6 \frac{u_0}{4!} \int^\Lambda \frac{d^d k_1}{(2\pi)^d} \int^\Lambda \frac{d^d k_2}{(2\pi)^d} \int^\Lambda \frac{d^d k_3}{(2\pi)^d} \int^\Lambda \frac{d^d k_4}{(2\pi)^d} \varphi_{\mathbf{k}_1}^< \varphi_{\mathbf{k}_2}^< \langle \varphi_{\mathbf{k}_3}^> \varphi_{\mathbf{k}_4}^> \rangle_{0,>} (2\pi)^d \delta^d \left(\sum_{i=1}^4 \mathbf{k}_i \right) \\ + \langle \text{terms with } \varphi^<(\varphi^>)^3, \varphi^>(\varphi^<)^3 \text{ and } (\varphi^>)^4 \rangle \end{aligned} \quad (95)$$

The factor of 6 is because of there are $\binom{4}{2}$ ways of choosing two $\varphi^<$ and two $\varphi^>$ out of four. Two of the terms that are not explicitly written with odd powers of $\varphi^>$ vanish, and $(\varphi^>)^4$ term gives a constant, which we are not considering presently. Now,

$$\langle \varphi_{\mathbf{k}_3}^> \varphi_{\mathbf{k}_4}^> \rangle_{0,>} = (2\pi)^d \delta^d(\mathbf{k}_3 + \mathbf{k}_4) \frac{1}{k_3^2 + r_0} \quad (96)$$

In deriving the above, we have taken the limits of the integral from $-\infty$ to ∞ , because, the major contribution to the gaussian comes from the region around zero. Using this,

$$\langle S_{\text{mix}} \rangle_{0,>} = 6 \frac{u_0}{4!} \int^\Lambda \frac{d^d k}{(2\pi)^d} \varphi_{\mathbf{k}}^< \varphi_{-\mathbf{k}}^< \int^\Lambda \frac{d^d q}{(2\pi)^d} \frac{1}{q^2 + r_0} \quad (97)$$

This goes into the correction of the gaussian.

$$r^< = r_0 + \frac{u_0}{2} \int^\Lambda \frac{d^d q}{(2\pi)^d} \frac{1}{q^2 + r_0} \quad (98)$$

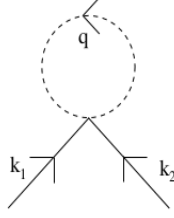
Since the correction is independent of $\mathbf{k}^2$, the coefficient of $(\nabla\varphi)^2$ term is zero. Hence $Z_b = 1$ upto order u_0 . If we go on to order u_0^2 , then the number of terms we have to look at become very large. Fortunately, there is a neat diagrammatic way of organizing these terms. The method of Feynman Diagrams. We give the rules here without deriving it⁴.

7.1 Feynman Rules

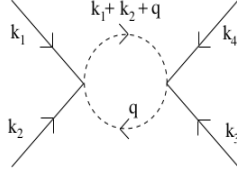
1. Each $\varphi_{\mathbf{k}}^<$ is represented by a solid line
2. $\varphi_{\mathbf{k}}^>$ is represented by a dotted line
3. Dotted lines connect to each other to form loops. No dotted line can be hanging loose.
4. Each factor of u_0 is represented by a vertex.
5. All diagrams conserve the total momentum $\mathbf{k}$
6. For each unknown loop momentum $\mathbf{q}$, add $\int^\Lambda \frac{d^d q}{(2\pi)^d}$
7. Each internal line is the propagator, $\langle \varphi_{\mathbf{k}}^> \varphi_{\mathbf{k}'}^> \rangle_{0,>}$
8. Each vertex is $u_0(2\pi)^d \delta^d(\text{sum of incoming momenta})$

⁴For derivation and more details, see [2, 4]

The rules tell us that a term of the form $u_0^n (\varphi^<)^m$ term in the effective action will be represented by a diagram which has n vertices and m solid lines. Thus, the term we have just calculated is,



To order u_0^2 , the correction to $(\varphi^<)^4$ comes from,



The diagram evaluates to,

$$2 \times 6^2 \times \left(\frac{u_0}{4!}\right)^2 \prod_i \left(\int_{\Lambda}^{\Lambda_0} \frac{d^d k_i}{(2\pi)^d} \varphi_{\mathbf{k}_i}^< \right) \int_{\Lambda}^{\Lambda_0} \frac{d^d q}{(2\pi)^d} \frac{1}{[\mathbf{q}^2 + r_0][(\mathbf{k}_1 + \mathbf{k}_2 + \mathbf{q})^2 + r_0]} \quad (99)$$

Since the momentum q lies within the shell $\Lambda < q < \Lambda_0$, we can safely set k_1 and k_2 to zero. Hence,

$$-\frac{1}{2} \left[\langle S_{\text{mix}}^2 \rangle_{0,>} - (\langle S_{\text{mix}} \rangle_{0,>})^2 \right] = -\frac{1}{2} \times 2 \times 6^2 \times \left(\frac{u_0}{4!}\right)^2 \prod_i \left(\int_{\Lambda}^{\Lambda_0} \frac{d^d k_i}{(2\pi)^d} \varphi_{\mathbf{k}_i}^< \right) \int_{\Lambda}^{\Lambda_0} \frac{d^d q}{(2\pi)^d} \frac{1}{[\mathbf{q}^2 + r_0]^2} \quad (100)$$

Hence,

$$u^< = u_0 - \frac{3u_0^2}{2} \int_{\Lambda}^{\Lambda_0} \frac{d^d q}{(2\pi)^d} \frac{1}{[\mathbf{q}^2 + r_0]^2} \quad (101)$$

Thus we have done the first step of Wilsonian RG perturbatively. Now, we rescale the field, and momenta.

$$\mathbf{k}' = b\mathbf{k} \quad \text{and} \quad \varphi'(\mathbf{k}') = b^{-1-d/2} \varphi(\mathbf{k}) \quad (102)$$

The quadratic term gives,

$$r' = b^2 Z_b r^< = b^2 \left[r_0 + \frac{u_0}{2} \int_{\Lambda}^{\Lambda_0} \frac{d^d q}{(2\pi)^d} \frac{1}{q^2 + r_0} \right] \quad (103)$$

Similarly, for the quartic term,

$$u' = b^{4-d} Z_b^2 u^< = b^{4-d} \left[u_0 - \frac{3u_0^2}{2} \int_{\Lambda}^{\Lambda_0} \frac{d^d q}{(2\pi)^d} \frac{1}{[\mathbf{q}^2 + r_0]^2} \right] \quad (104)$$

We have used the fact that $Z_b = 1$ in this order.

8 The Wilson-Fisher Fixed Point [2, 4, 5]

To form the differential equations, let,

$$b = e^\ell \quad (105)$$

Denote $r_\ell = r'$ and $u_\ell = u'$. Now,

$$\frac{\partial}{\partial \ell} \int_{\Lambda}^{\Lambda_0} \frac{d^d q}{(2\pi)^d} \frac{1}{q^2 + r_0} = \Omega_d \frac{\partial}{\partial \ell} \int_{\Lambda=\Lambda_0 e^\ell}^{\Lambda_0} \frac{dk k^{d-1}}{r_0 + k^2} \sim \frac{\Omega_d \Lambda^d}{r_0 + \Lambda^2} \quad (106)$$

This gives,

$$\frac{\partial r_\ell}{\partial \ell} = 2r_\ell + e^{2\ell} \frac{u_0}{2} \frac{\Omega_d \Lambda^d}{r_0 + \Lambda^2} \quad (107)$$

If we are near the critical point, and assume ℓ is small, then, approximate $r_0 = r_\ell$, $u_0 = u_\ell$

$$\frac{\partial r_\ell}{\partial \ell} = 2r_\ell + \frac{u_\ell}{2} \frac{\Omega_d \Lambda_0^d}{r_\ell + \Lambda_0^2} \quad (108)$$

Likewise for small ℓ ,

$$\frac{\partial u_\ell}{\partial \ell} = (4-d)u_\ell - \frac{3u_\ell^2}{2} \frac{\Omega_d \Lambda_0^d}{(r_\ell + \Lambda_0^2)^2} \quad (109)$$

Let,

$$\tilde{r}_\ell = \frac{r_\ell}{\Lambda_0^2} \quad \text{and} \quad \tilde{u}_\ell = \frac{\Omega_d u_\ell}{\Lambda_0^{4-d}} \quad (110)$$

Then,

$$\frac{\partial \tilde{r}_\ell}{\partial \ell} = 2\tilde{r}_\ell + \frac{\tilde{u}_\ell}{2(1 + \tilde{r}_\ell)} \quad (111)$$

and

$$\frac{\partial \tilde{u}_\ell}{\partial \ell} = (4-d)\tilde{u}_\ell - \frac{3\tilde{u}_\ell^2}{2(1 + \tilde{r}_\ell)^2} \quad (112)$$

It is known that above four dimensions, the mean field approximation of the ising model breaks down below 4 dimensions. We choose d such that[5],

$$\epsilon = 4 - d \quad (113)$$

is small. The fixed points of the above differential equations are given by,

$$2\tilde{r}_* + \frac{\tilde{u}_*}{2(1 + \tilde{r}_*)} = 0 \quad \text{and} \quad (4-d)\tilde{u}_* - \frac{3\tilde{u}_*^2}{2(1 + \tilde{r}_*)^2} = 0 \quad (114)$$

To get solutions of order ϵ ,

$$\frac{3\tilde{u}_*^2}{2(1 + \tilde{r}_*)^2} \times \frac{2(1 + \tilde{r}_*)^2}{3\tilde{u}_*} = \epsilon \tilde{u}_* \times \frac{2(1 + \tilde{r}_*)^2}{3\tilde{u}_*} \quad (115)$$

Hence,

$$\tilde{u}_* = \frac{2}{3}(1 + \tilde{r}_*)^2 \epsilon + \mathcal{O}(\epsilon^2) \quad (116)$$

Using this in the equation for $\tilde{r}_*$,

$$\tilde{r}_* = -\frac{1}{4}(1 + \tilde{r}_*)\epsilon + \mathcal{O}(\epsilon^2) \quad (117)$$

Keeping terms upto $\mathcal{O}(\epsilon^2)$,

$$\tilde{u}_* = \frac{2}{3}\epsilon + \mathcal{O}(\epsilon^2) \quad \text{and} \quad \tilde{r}_* = -\frac{1}{6}\epsilon + \mathcal{O}(\epsilon^2) \quad (118)$$

The above fixed points are called the Wilson-Fisher Fixed point. One can show that at this fixed point, the correlation length is infinite, hence this is a critical point. To find the critical exponents, we linearise the flow

around the critical point. Let δr_ℓ and δu_ℓ be changes from the critical point. Then the linearised flow equation becomes,

$$\partial_\ell \begin{pmatrix} \delta r_\ell \\ \delta u_\ell \end{pmatrix} = \begin{pmatrix} 2 - \frac{\epsilon}{3} & \frac{1}{2} + \frac{\epsilon}{12} \\ 0 & -\epsilon \end{pmatrix} \begin{pmatrix} \delta r_\ell \\ \delta u_\ell \end{pmatrix} \quad (119)$$

The eigenvalues of the linearised RG flow matrix are, $2 - \frac{\epsilon}{3}$ and $-\epsilon$. One positive, and one negative. From scaling hypothesis, it is known that the relevant scaling exponent is directly related to the critical exponent ν as,

$$\nu = \frac{1}{2 - \frac{\epsilon}{3}} = \frac{1}{2} + \frac{\epsilon}{12} + \mathcal{O}(\epsilon^2) \quad (120)$$

Since the critical exponent η is related to the anomalous dimension Z_b and it is 1 upto this order,

$$\eta = 0 + \mathcal{O}(\epsilon^2) \quad (121)$$

Now,

$$\alpha = 2 - \nu d = 2 - \left(\frac{1}{2} + \frac{\epsilon}{12} \right) (4 - \epsilon) + \mathcal{O}(\epsilon^2) = \frac{\epsilon}{6} + \mathcal{O}(\epsilon^2) \quad (122)$$

Similarly,

$$\beta = \frac{(d - 2 + \eta)\nu}{2} = \frac{1}{2} - \frac{\epsilon}{6} + \mathcal{O}(\epsilon^2) \quad (123)$$

$$\gamma = \nu(2 - \eta) = 1 + \frac{\epsilon}{6} + \mathcal{O}(\epsilon^2) \quad (124)$$

$$\delta = \frac{d + 2 - \eta}{d - 2 + \eta} = 3 + \epsilon + \mathcal{O}(\epsilon^2) \quad (125)$$

Now, we do something rather extreme. Put $\epsilon = 1$, and compare the results with experiments. Here are they,

	α	β	γ	δ	η	ν
MFT	0	0.5	1	3	0	0.5
RG	0.17	0.33	1.17	4	0	0.58
Exptal	0.1101	0.3264	1.2371	4.7898	0.0363	0.6300

Even with this extreme usage of the ϵ expansion, we get pretty close to the actual values!!

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